

Surender Singh

surenderchoudhary07072004@gmail.com | github.com/Surender707 | linkedin.com/in/surender-choudhary-2a2b46293

PROFESSIONAL SUMMARY

Results-driven AI & Data Science engineer with hands-on expertise in building and deploying end-to-end machine learning systems spanning Computer Vision, NLP, and predictive analytics. Proven ability to bridge the gap between research-grade AI models and production-ready full-stack applications using modern MLOps and web technologies. Actively seeking a Machine Learning Engineer or Data Scientist role to deliver scalable, intelligent solutions.

EDUCATION

B.Tech in Artificial Intelligence & Data Science (AI&DS) | CGPA: 7.58
Modern Institute of Technology and Research Center (MITRC) — 2023 – Present (Expected 2027)

TECHNICAL SKILLS

Languages	Python (Expert), JavaScript, C++, Java, SQL, HTML/CSS
AI / ML / Deep Learning	PyTorch, TensorFlow, scikit-learn, LSTM, CNN, HuggingFace Transformers
Computer Vision & NLP	YOLOv8, OpenCV, MediaPipe, PyAutoGUI
Data & Analytics	Pandas, NumPy, Gradient Boosting, EDA, Feature Engineering
Full-Stack & Deployment	Flask, React, Node.js, Express.js, MongoDB, MySQL, JWT, Chart.js
DevOps & Tools	Git/GitHub, Docker

PROFESSIONAL EXPERIENCE

Machine Learning Intern | CodSoft | Remote | 2024

- Engineered an end-to-end **Credit Card Fraud Detection** model with advanced feature engineering and class-imbalance handling (SMOTE), achieving high-precision fraud identification on real-world skewed datasets.
- Developed a **Titanic Survival Prediction** pipeline with full EDA, preprocessing, and ensemble classification, demonstrating proficiency across the complete ML lifecycle.
- Built a **Movie Rating Prediction** regression model using feature encoding and hyperparameter tuning, delivering accurate rating forecasts across varied film metadata.

Data Science Intern | Saiket | Remote | 2024

- Architected a full **Customer Churn Analysis** pipeline — spanning data cleaning, EDA, feature selection, predictive modeling, and business-level evaluation — from raw CRM data to actionable insights.
- Identified key churn drivers and delivered a structured report enabling data-backed retention strategy recommendations for stakeholders.

PROJECTS

TaskFlow | MERN Stack, Node.js, MongoDB, Chart.js

- Built an intelligent full-stack productivity system featuring a custom **Intelligence Engine** for automated task prioritization and behavioral recommendations.
- Integrated **Chart.js** dashboards for real-time productivity metric tracking and task completion visualization.

PlayStore Pulse | Python, Flask, Gradient Boosting, React

- Developed an analytics platform processing **10,000+ Play Store app records** with interactive dashboards for ratings, installs, and category insights.
- Trained and deployed a **Gradient Boosting rating predictor** via a Flask REST API — demonstrating full end-to-end ML deployment from training to live API consumption.

Emotion Classifier | YOLOv8, PyTorch, CNN, OpenCV

- Built a real-time facial emotion recognition system combining **YOLOv8** face detection with a custom-trained **CNN (PyTorch)** classifying across **7 emotion categories** with live webcam inference.

Hand Gesture Virtual Mouse | MediaPipe, OpenCV, PyAutoGUI

- Engineered a touchless HCI system using **MediaPipe hand landmark tracking** to map finger gestures to system controls (clicks, drags, zoom) — eliminating physical input devices.

Dual-Model Text Summarizer | HuggingFace Transformers, LSTM, Attention, PyTorch

- Designed a comparative NLP tool implementing a custom **LSTM with Attention** (trained from scratch) alongside a fine-tuned **HuggingFace T5 transformer** for side-by-side abstractive summarization benchmarking.

CERTIFICATIONS

- Artificial Intelligence Journey: Beginner to Pro — Udemy, Data Science Academy & School of AI (Apr 2026)
- Machine Learning & AI Foundations Course — Udemy (Apr 2026)
- Certified Master in Artificial General Intelligence Systems — Udemy School of AI (Apr 2026)
- Python Mastery: 100 Days, 100 Projects — Udemy School of AI (Feb 2026)
- Google Generative AI Leader Exam: Complete Preparation Guide — Udemy (Jan 2026)
- Ultimate DevSecOps Bootcamp — Udemy School of DevOps (Feb 2026)
- Agentic AI Playbook: Complete Guide for Tech Leaders — Udemy School of DevOps (Feb 2026)

ACHIEVEMENTS

- **1st Prize — College Hackathon:** Demonstrated exceptional problem-solving and rapid prototyping skills under time constraints, outperforming competing teams in a competitive technical challenge.